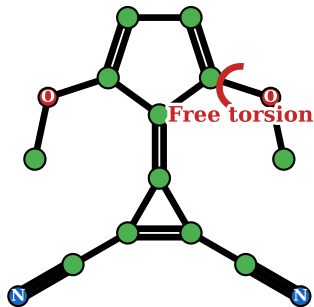
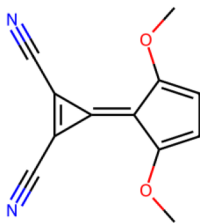


(a) 3D Conformer



Torsional degrees of freedom visible

(b) 2D Molecular Graph



Topological connectivity only — no spatial information

(c) Representation Bottleneck

Fed. 3D SchNet
(Rigid D-A, n=13)

$R^2=+0.852$

Fed. 3D SchNet
(E14, n=53)

*Conformational
fidelity gain*

$R^2=+0.181$

*3D repr.
gain*

Fed. 2D GIN
(E8, n=53)

$R^2=+0.028$

*Federation
gain*

Local 2D GIN
(E5, n=53)

$R^2=-0.677$

